Air Quality Sensor Data Analysis

COMP5048 Assignment 2

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I. Task 1

A. Data and Preprocessing

We used the Air Quality dataset from the UCI repository [1]. It contains hourly averaged responses of a metal-oxide multisensor device placed at road level in an Italian city, together with reference analyser values and weather readings [2]. The file has 9,357 hourly records from 10 March 2004 18:00 to 4 April 2005 14:00, with no duplicate or missing timestamps, and 13 value columns. Missing values are coded as −200.

Before cleaning, we checked the raw file ourselves. The main findings were:

- −200 is the only missing code. CO(GT) is 18.0% missing, NOx(GT) and NO2(GT) 17.5%, NMHC(GT) 90.2% (valid only until 1 May 2004) and every other column 3.9%.
- Missing values are not random. All sensor columns, C6H6, T, RH and AH go missing together (366 hours), 31 hours have no data at all, NOx and NO2 are missing at 03:00 on 93% of days and CO at 04:00 on 54% of days.
- C6H6(GT) is almost a perfect function of PT08.S2(NMHC) (Spearman 0.99998): the same S2 value always gives the same C6H6 value within 0.1.
- AH matches the water vapour pressure (kPa) calculated from T and RH, so it is a derived column. NOx(GT) is in ppb, while NO2(GT) is in µg/m³.
- The sensors drift, as reported in [2]. At a fixed NOx level, the PT08.S3 response ranges from about 620 to 923 depending on the month, and PT08.S4 follows T and AH (r = 0.56 and 0.63) more than NO2 (r = 0.16).

Our cleaning script reads the CSV (semicolon separator, decimal comma), drops 114 blank rows and two empty columns, builds one hourly timestamp and replaces −200 with NaN. Nothing is imputed. The script then checks that no rows were lost and no −200 remains, that the number of NaNs in each column equals the number of −200 values in the raw file, and that all other values are unchanged.

B. Attribute Selection

We selected the timestamp, CO(GT), C6H6(GT), NOx(GT), NO2(GT), PT08.S1(CO), PT08.S5(O3), T and RH (nine attributes) and used them in every task. This covers true concentrations, sensor responses and the environment. We left out NMHC(GT) because it is mostly missing, AH because it is derived from T and RH, PT08.S2 because it duplicates C6H6, and PT08.S3 and PT08.S4 because they showed the strongest drift and temperature effects. Views that show all attributes at once need complete hours, which gives 6,941 hours (74%). October 2004 is the weakest month, at 50%.

Some attributes overlap. C6H6 and CO are strongly related (r = 0.93), and S1 and S5 both follow CO (r = 0.88 and 0.85). A group defined by high CO, C6H6, S1 and S5 is therefore one traffic signal seen several times, not four separate factors, and we describe the groups with this in mind.

C. Visual Analytic Tool

We built our own web application with four linked views. All views use the same 6,941 complete hours, and the hour of day, weekday, month and season are derived from the timestamp.

- Parallel coordinates [3] is the main view. Each hour is one line, so hours with similar profiles form bundles, and the axes a bundle passes through show which attributes define it. Lines can be coloured by season, time of day (four 6-hour bands), weekday/weekend or saved group.
- A scatterplot of any two attributes shows separation and density for a pair.
- A daily heatmap, with one row per day, shows the daily median of each attribute by colour lightness and cell width, so groups of days stand out.
- A group panel shows, for each hour of day, weekday and month, the share of hours in that slot that belong to a group, together with the group medians.

The axes are ordered analysers, then sensors, then T and RH, so related pollutants sit next to each other and bundles appear as parallel lines. Axes can be dragged to reorder them. Each axis spans the 1st to 99th percentile with its raw end values labelled, because with the full range a few extreme hours pushed most lines to the bottom. The ranges stay fixed when filtering, so views can be compared. With almost 7,000 lines we used low opacity and a fixed random drawing order, so that no season is always drawn on top, and clicking a legend entry shows only that category's lines. We checked the colours for colour-blind safety; only three passed every pair, so the tool allows three saved groups.

Following "overview first, zoom and filter, then details-on-demand" [4], the user starts from all hours, filters by date, season, month, day type and hour, and then selects hours by brushing ranges on one or more axes, dragging a box in the scatterplot or clicking days in the heatmap. The selection is highlighted in all views and can be saved as a group with its own colour and a thick median line. No clustering algorithm was used; the groups were found only by looking at the views and brushing.

D. *Discovery of Groupings*

We first coloured the parallel coordinates by season and showed one season at a time (Fig. 1). Winter lines sit at the bottom of T and the top of RH, and many reach high on NO2. Summer lines meet at the top of T and the bottom of RH, and drop to the floor of the NOx axis. Autumn held most of the lines running along the top of every pollutant and sensor axis. Colouring by time of day showed night lines (00–05) along the bottom of all pollutant axes and afternoon lines (12–17) in the hot, dry bundle.

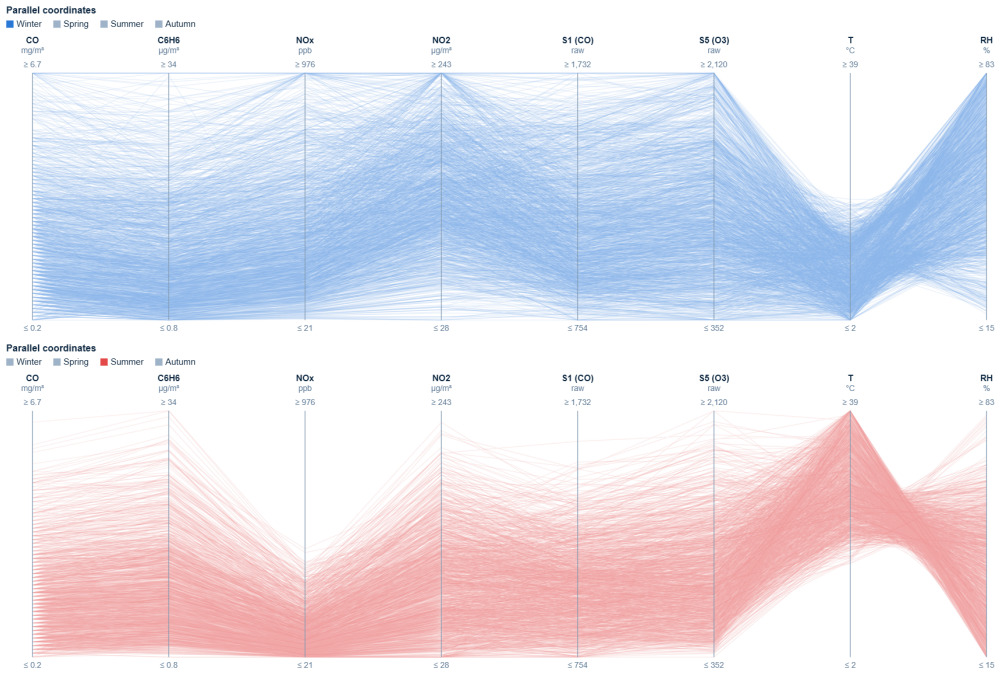


Fig. 1. Parallel coordinates showing only winter (top) and only summer (bottom).

We then brushed each bundle where it sits on the axes:

- Group A: the upper half of the CO axis (CO ≥ 3.5 mg/m³), where lines also stay high on the other pollutant and sensor axes;
- Group B: the bottom band of CO and NOx (CO ≤ 0.9 mg/m³ and NOx ≤ 90 ppb), where the night lines are;
- Group C: the top third of T and the bottom of RH (T ≥ 26 °C and RH ≤ 35%);
- Group D: the lower part of T (T ≤ 10 °C).

A, B and C were saved as groups (Fig. 2), and D was kept as the current selection because the tool allows three saved groups. Table I summarises the groups using the group panel (Fig. 3). The percentages are the bar heights in that panel; for example, 36% at 08:00 means that 108 of the 303 complete hours at 08:00 belong to Group A.

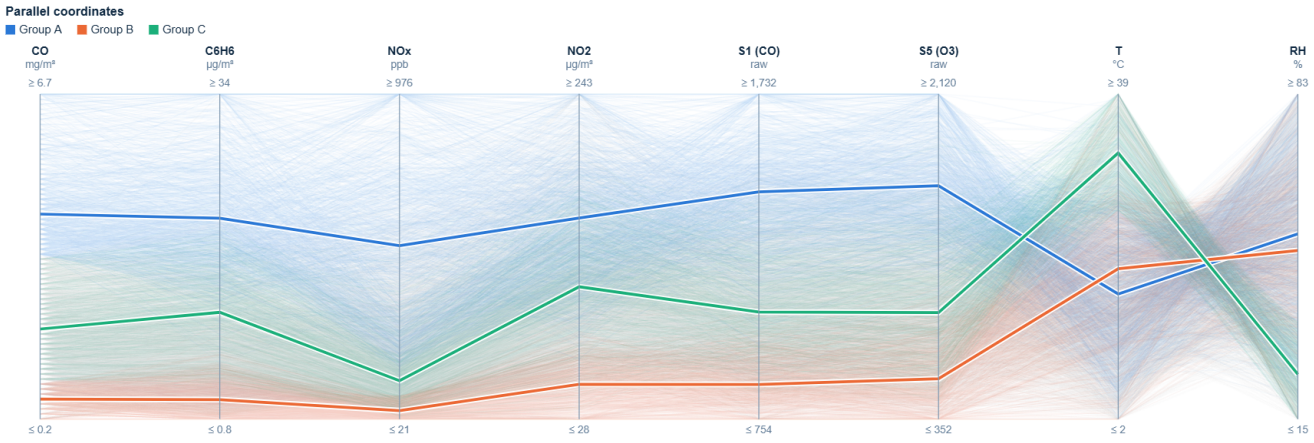


Fig. 2. Parallel coordinates coloured by saved group. Thick lines are the group medians.

TABLE I. GROUPS FOUND IN THE PARALLEL COORDINATES

Group	Brush	Hours	Defining attributes	Time periods
A: traffic peaks	CO ≥ 3.5	1,172 (17%)	High on all pollutant and sensor axes. Medians CO 4.3, C6H6 22, NOx 531, NO2 161 (all hours: 1.9, 8.8, 186, 110). T and RH typical	08–09 (36–37%) and 18–20 (44–51%). Tue–Fri 22–24%, Mon 14%, Sat 9%, Sun 4%. Almost none in Aug 2004 (1 hour); highest Oct–Nov 2004 (29–31%)
B: clean nights	CO ≤ 0.9, NOx ≤ 90	855 (12%)	Floor of all pollutant and sensor axes. Medians CO 0.6, C6H6 2.8, NOx 46, NO2 51	02–06 (64% at 04–05). Sun 24% vs 8–14% on other days. Mostly Apr–Aug 2004 (16–29%); rare Oct 2004–Mar 2005 (4–7%)
C: hot, dry afternoons	T ≥ 26, RH ≤ 35	906 (13%)	High T (median 33 °C), low RH (24%) and low NOx (134); CO and NO2 typical	10:00–20:00 (28–32% at 12–17). All days 11–16%. May–Sep 2004, peak Jul 2004 (53%); none Oct 2004–Feb 2005
D: cold, humid hours	T ≤ 10	1,431 (21%)	Low T (6.6 °C), higher RH (57%), high NOx (254) and NO2 (131) but CO below typical (1.6)	Mostly 00–09 (25–34%, except 03:00) vs 9–11% at 12–17. All days similar. Mainly Nov 2004–Mar 2005, peak Feb 2005 (78%)

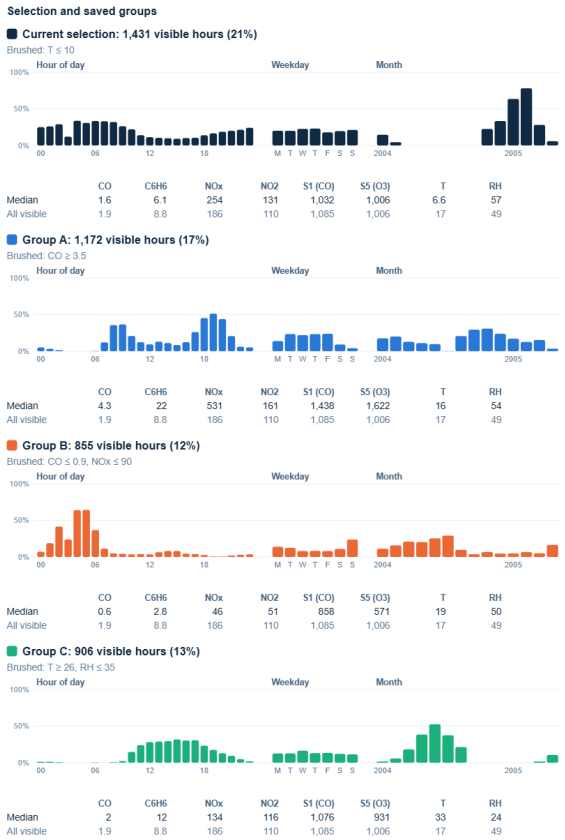


Fig. 3. Group panel: share of hours in each hour of day, weekday and month that belong to each group, with medians. The top card is the cold-hours selection (Group D).

E. *Groupings over Hours and Days*

The groups depend on the hour and the day, and the same hour can belong to different groups. Afternoons are often in Group C in summer but never in winter, and early mornings are often in Group B in spring and summer but rarely in winter (Fig. 3). The daily heatmap also shows groups of days (Fig. 5). In November 2004 the Sundays (7, 14 and 21 November) are lighter on every pollutant column and have almost no Group A hours, while 22–27 November forms a dark block that looks like a pollution episode lasting several days.

In the scatterplot (Fig. 4), Groups A and B sit at opposite ends of the CO–C6H6 band, while Group C sits on its upper edge, so hot hours have more C6H6 for the same CO. Because C6H6 comes from a temperature-sensitive sensor, we think this is at least partly a sensor effect.

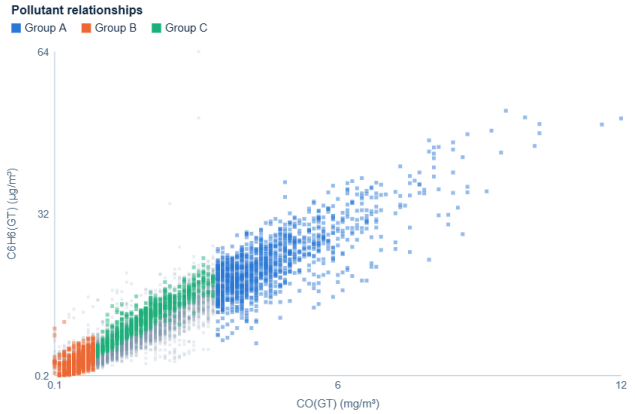


Fig. 4. CO against C6H6, coloured by group.

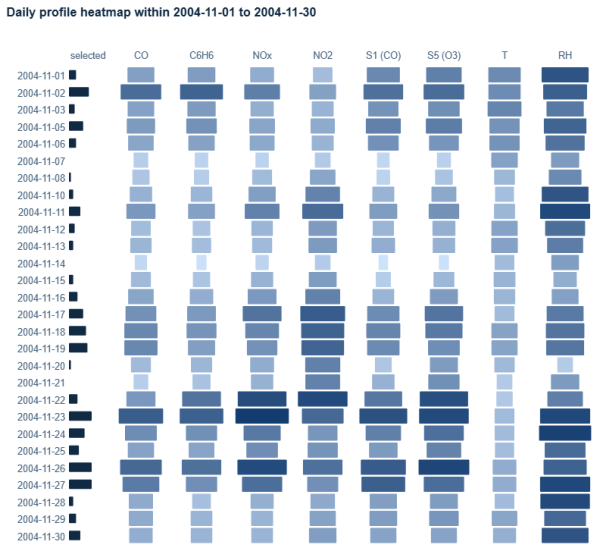


Fig. 5. Daily heatmap for November 2004 with the CO ≥ 3.5 brush active. The left bar counts high-CO hours per day.

F. *Robustness of the Brushes*

The brush positions are our visual choices, not exact boundaries, because the data is continuous. We moved each brush in both directions and read the group panel again (Table II). The group sizes and percentages change, but the patterns stay the same. A always has the morning and evening peaks and almost no hours at night, on Sundays or in August; B is always night-time and more common on Sundays; C is always a summer daytime group; and D is always a winter group concentrated at night and in the morning.

TABLE II. EFFECT OF MOVING EACH BRUSH (LOWER / USED / HIGHER)

Group	Brush	Hours	Share of hours in selected slots
A	CO ≥ 3.0 / 3.5 / 4.0	1,692 / 1,172 / 814	08:00: 49 / 36 / 26%; 19:00: 63 / 51 / 41%; Sunday: 7 / 4 / 3%; 04:00 and Aug 2004: 0% in all
B	CO ≤ 0.7, NOx ≤ 70 / 0.9, 90 / 1.1, 110	553 / 855 / 1,231	04:00: 51 / 64 / 74%; 19:00: 0 / 1 / 3%; Sunday: 16 / 24 / 35% vs Wednesday: 5 / 8 / 11%
C	T ≥ 24, RH ≤ 40 / 26, 35 / 28, 30	1,221 / 906 / 646	14:00: 36 / 30 / 24%; 04:00: 1 / 0 / 0%; Jul 2004: 64 / 53 / 42%; Nov 2004 and Feb 2005: 0% in all
D	T ≤ 8 / 10 / 12	999 / 1,431 / 1,989	04:00: 28 / 34 / 47% vs 14:00: 5 / 10 / 16%; Feb 2005: 61 / 78 / 90%; Jul 2004: 0% in all

G. Limitations

Groups defined by high CO and C6H6 reflect one traffic signal, and C6H6 depends on a temperature-sensitive sensor. The sensors also drift, so seasonal differences on the S1 and S5 axes are partly instrumental. The groups can overlap: A and C share 102 hours and B and C share 97, while A and B do not overlap. Finally, because only complete hours are used, months with analyser gaps such as October 2004 have fewer hours, and the 03:00 slot has very few complete hours.

II. TASK 2

A. Temporal Views

To follow the groups over time, we added three linked views to the same tool. They use the groups from Task 1; a "Restore Task 1 groups" button recreates them exactly, with Group D as the current selection.

- A membership matrix for each group, with months as rows and hours of day as columns. The colour of a cell is the share of complete hours in that month and hour that belong to the group, from light grey (0%) to the group colour (100%). Hatched cells have no complete hours and a dot marks cells with fewer than five. Reading down a column shows one hour of day changing group over the year.
- A monthly profile for each group, with months as rows and attributes as columns. Red indicates that the group's median is higher than the median of all visible hours in the same month, including the group itself; blue indicates a lower median, and grey indicates no difference. The numbers represent the group medians. Comparing within each month removes the seasonal level, so the colours show what makes the group different at that time. Months with fewer than five group hours are left blank.
- Monthly parallel coordinates: one small plot per month with the same axes, group lines and group medians, so the shape of each bundle can be compared month by month as small multiples [5].

The matrix keeps both time scales visible at once: the daily cycle across and the seasons down. The profile uses a diverging blue–grey–red scale because a difference has a natural zero. All views use the same axis order and scaling as Task 1, so a pattern seen in one view can be checked in the others.

B. Changes in Group Membership

Fig. 6 shows that the membership of every group changes during the year:

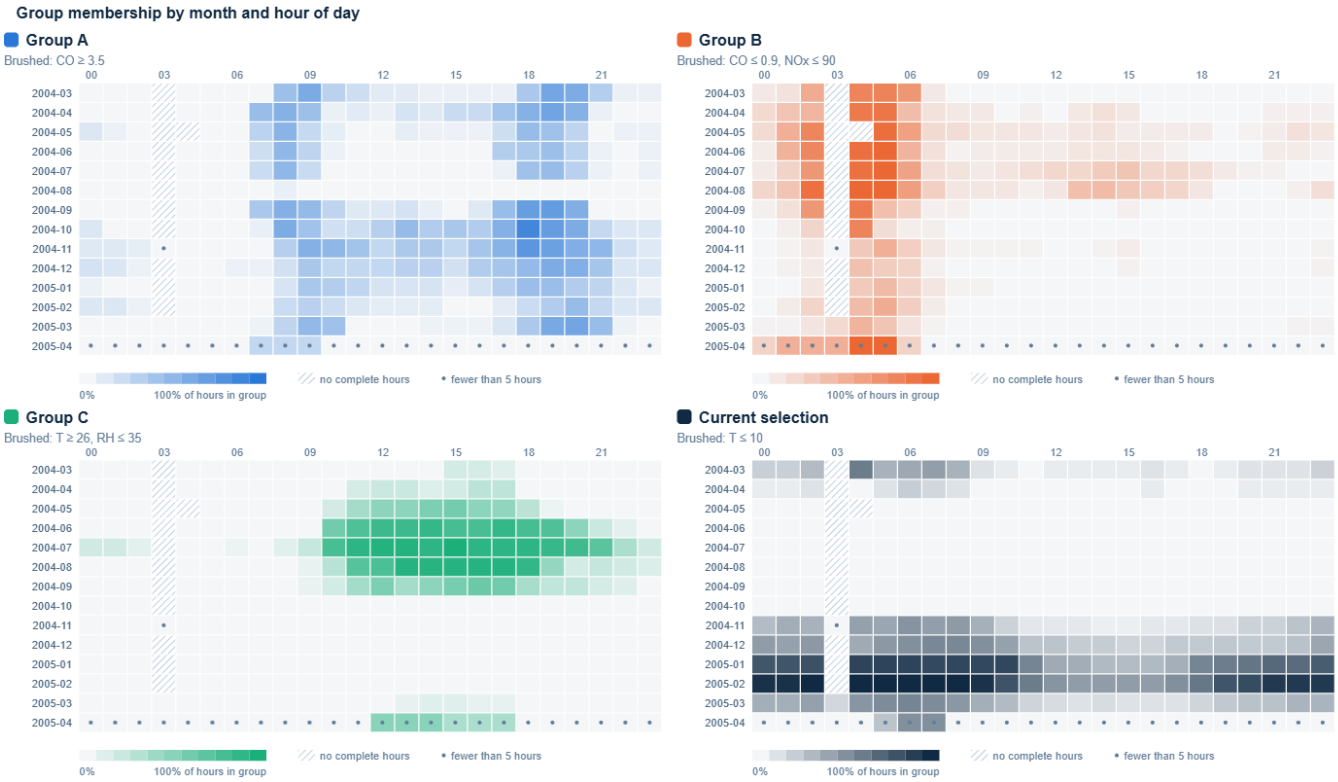


Fig. 6. Membership matrices (month × hour of day) for Groups A, B and C and the cold-hours selection (D).

- Group A: from March to July 2004 only the two rush-hour peaks belong to A. In August 2004 the group almost disappears (1 of 408 hours). From November to February midday hours join it (11:00–16:00 rises from 5% to 20%) and so do late-night hours (00:00–02:00, from 1% to 8%), while the rush-hour peaks hardly change (43% in spring and summer, 42% in winter). By March 2005 the midday hours have left again (2%).
- Group B: its core is 04:00–05:00 in every month, but 99% of those hours belong to B in July–August against 40% in November–February. In July–August daytime hours also join (23% of 10:00–17:00, against 1% in winter).
- Group C: in April–May it covers midday only (38% of 12:00–16:00, almost nothing after 19:00). In June–August it widens to 77% of 10:00–21:00. It is absent from October to February and returns at midday in March–April 2005.
- Group D: it covers 71% of all hours in January–February, but in March–April 2004 only nights and mornings (21% of 00:00–08:00 against 3% of 12:00–17:00).

C. *Changes in the Defining Attributes*

The monthly profiles (Figs. 7–9) show that the combination of attributes that makes a group different also changes:

- Group A: from March to July it is mainly a CO, C6H6 and S1 group, with NOx only moderately raised (median 227–296 ppb). From September NOx becomes much more important (506–733 ppb, darker red), and NO2 and S5 also rise, with the highest NO2 median in February 2005 (221 µg/m³). In the monthly parallel coordinates (Fig. 10) the median of A dips at the NOx axis in spring and summer but stays high across NOx in autumn and winter. In June and July, Group A also has lower RH than the monthly median across all visible hours.
- Group B: from March to September it is defined mainly by lower CO, C6H6 and S1, while NOx and NO2 are only a little lower than usual. In August 2004, the row is almost grey, indicating that Group B's medians are close to the overall monthly medians; this matches B spreading into the daytime that month. From November to February the strongest blue is on NOx, NO2 and S5, so in winter the clean hours are defined by low nitrogen oxides rather than CO.
- Group C: in March–April of both years it is strongly defined by high T and low RH, which marks rare warm, dry spells. In June–August these differences fade because all hours are warm, and in September higher NOx and NO2 (329 ppb and 145 µg/m³) join the profile.

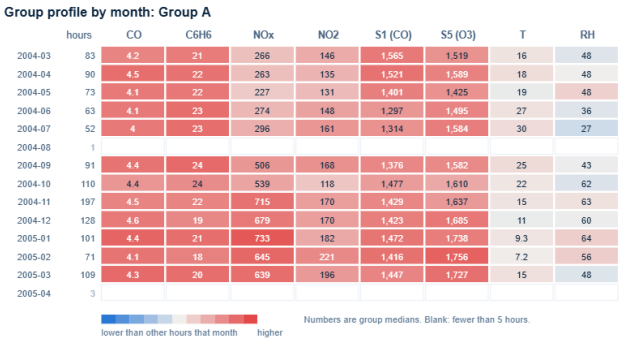


Fig. 7. Monthly profile of Group A.

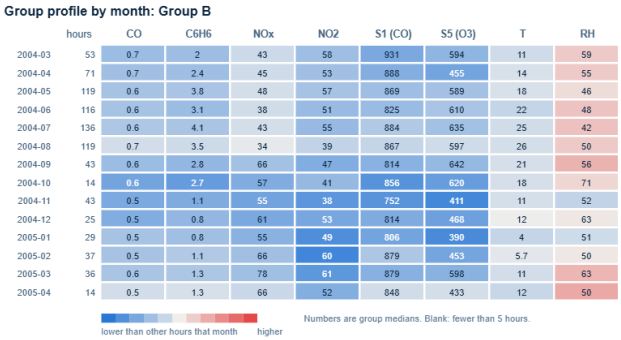


Fig. 8. Monthly profile of Group B.

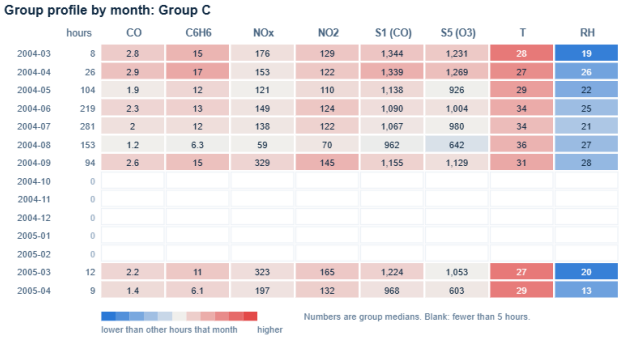


Fig. 9. Monthly profile of Group C.

For Group D, low T and high RH define the group in November and March–April 2004. However, in January–February, Group D covers most hours, so its medians are close to the overall monthly medians.

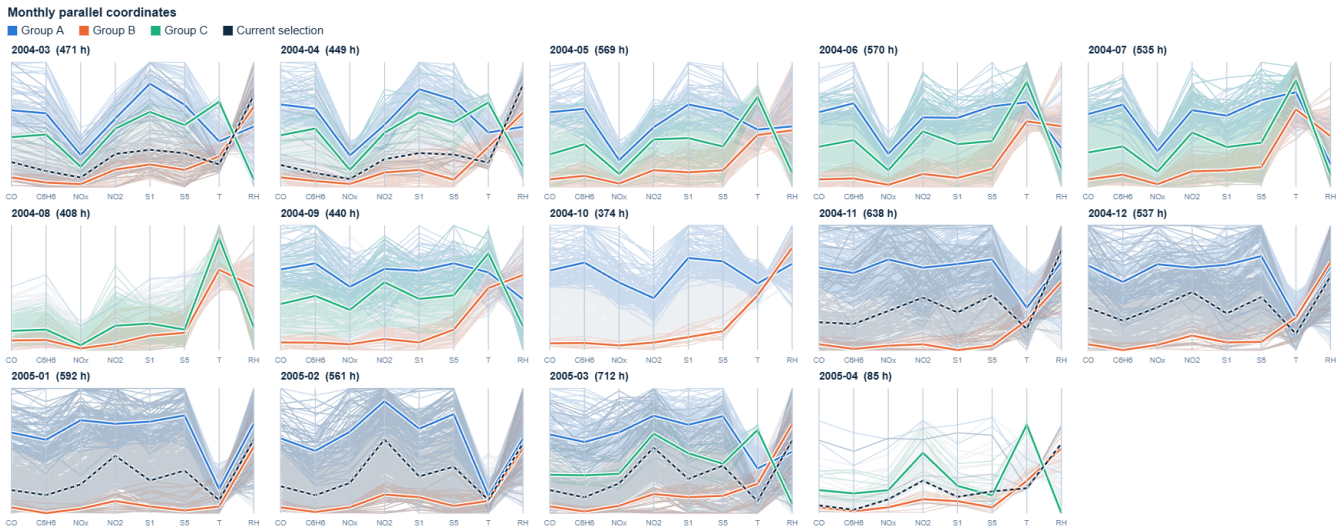


Fig. 10. Monthly parallel coordinates. Thick lines are group medians for each month; the dashed line is Group D.

D. Summary and Limitations

Both the membership and the defining combination change over the year. The biggest change is between summer and autumn 2004: in August Group A almost disappears and Group B spreads into the daytime, and from September Group A becomes led by NOx and starts to take in midday hours. Task 3 looks at this period. Some of the change on the sensor axes (S1, S5) may come from sensor drift. Months with long analyser gaps, such as October 2004, and April 2005, which has only 85 complete hours, are based on fewer hours.

III. Task 3

A. Identifying the Critical Moment

Task 2 showed that Group C ($T \geq 26$ °C and RH $\leq 35\%$) disappears in autumn: the membership matrix (Fig. 6) has 94 Group C hours in September 2004 and none in October. To find when this happens, we used interactive date–hour heatmaps of Group C membership, temperature and relative humidity for 25 September–10 October, 11:00–17:00 (Fig. 11). The three panels share the same rows and columns, so the same hour can be compared directly, and hovering a cell shows its values. Grey cells are hours with missing data.

The heatmaps show a clear break between 29 and 30 September. At 14:00–17:00, all four hours belong to Group C on 29 September and none on 30 September, and all eight hours are available. 29 September at 17:00 is the last Group C hour of 2004; the group only returns on 16 March 2005.

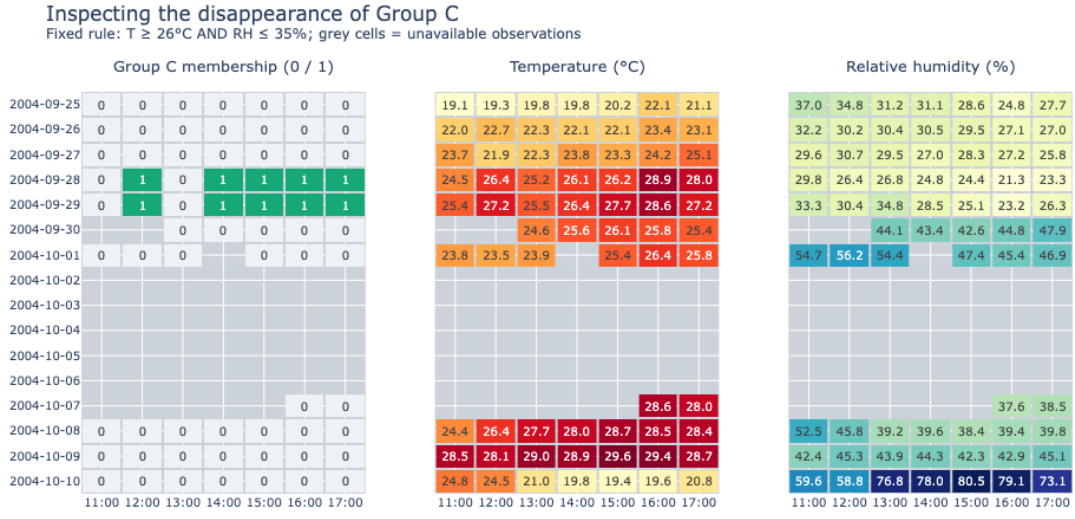


Fig. 11. Interactive date–hour heatmaps of Group C membership, temperature and relative humidity, 25 September–10 October 2004, 11:00–17:00. Grey cells are missing hours.

B. What Changed

On 29 September the four hours have temperatures of 26.4–28.6 °C and relative humidity of 23.2–28.5%. On 30 September the humidity rises to 42.6–47.9% at every hour and the temperature falls below 26 °C at 14:00, 16:00 and 17:00 (26.1 °C at 15:00), so all four hours leave Group C.

The pollutants change much less. CO is 2.4, 2.8, 3.6 and 4.0 mg/m³ on 29 September and 2.2, 2.8, 3.9 and 6.5 mg/m³ on 30 September. The 16:00 and 17:00 hours are in Group A

on both days, so they move from overlapping Groups A and C to Group A only, while 14:00 and 15:00 move from Group C to no group. The change is therefore in the environmental part of the profile, while the traffic pattern stays the same.

C. What-if Scenarios

We tested three alternatives for the 30 September hours, each using the previous day's value at the same hour: temperature only, humidity only, and both together. Everything else, including the pollutant values and the group rules, is unchanged. The scenario heatmap in the tool shows the result for each hour (Fig. 12):

- Restoring temperature only keeps 0/4 hours in Group C, because humidity stays above 35%.
- Restoring humidity only keeps 1/4 (15:00, where the temperature already meets the rule).
- Restoring both keeps 4/4.

So the membership loss would have been avoided if both temperature and humidity had stayed at the previous day's level; changing only one of them is not enough. Because CO is unchanged, the 16:00 and 17:00 hours also stay in Group A, so all four hours keep exactly the same groups as on 29 September.



Fig. 12. What-if scenarios for 30 September 2004, 14:00–17:00, using the previous day's temperature and/or humidity. Only the joint scenario keeps all four hours in Group C.

D. Limitations

Because Group C is defined by temperature and humidity, this result follows directly from the group rule; it does not show that the weather changes the pollution. We also tried a model-based what-if for a Group A change at 17:00 on 20 September 2004, using Random Forest models with temperature, humidity and time of day. Adding temperature and humidity did not improve the models (test R^2 0.404 against 0.418 for CO, and 0.145 against 0.280 for NOx), and the CO model predicted 2.63 mg/m³ against the observed 5.0 mg/m³, so we did not use it to claim that environmental changes would prevent entry into Group A. The full analysis is in the Task 3 notebook.

IV. Task 4

A. Evaluation Method

We conducted a task-based usability evaluation to assess whether the visual analytic tool supported the analytical objectives of Tasks 1–3. The evaluation involved one university student who was unfamiliar with the dashboard and did not have a background in data science or computer science.

Four tasks were designed: E1 required discovering and saving groups using linked visualisations; E2 involved comparing monthly group membership and attribute profiles; E3 focused on identifying changes in Group C using date–hour heatmaps; and E4 required interpreting the predefined what-if scenarios.

The evaluation focused on task completion, difficulties encountered during interaction, and the participant's interpretation of visual evidence.

B. Evaluation Findings

The evaluation session revealed several key usability issues and interaction bottlenecks across the four tasks.

In E1 (Group Discovery), the participant required verbal guidance to interact with the parallel coordinates plot. Because the brushing mechanism lacked intuitive visual signifiers or prompts, the participant did not immediately realize that axes could be brushed, hindering their ability to identify and save clusters independently. Once the interaction was demonstrated, the participant successfully selected and saved the target groups.

In E2 (Temporal Analysis), the participant easily identified overall shifts in monthly group membership but encountered significant confusion when interpreting the monthly attribute heatmap. Specifically, the participant exhibited a cognitive mismatch regarding the color encoding, misinterpreting the red hues as absolute indicators of "severe pollution" rather than values exceeding the relative monthly baseline. The participant explicitly questioned the underlying baseline reference, highlighting ambiguity in the visual scale.

In E3 (Critical Moment Analysis), the participant initially conflated cells representing missing data with time intervals that simply fell outside Group C. After reviewing the legend and clarifying the encoding schema, the participant successfully conducted cross-heatmap comparisons, linking corresponding dates and hours across group membership, ambient temperature, and relative humidity.

In E4 (What-if Analysis), the participant struggled to distinguish between the predefined rule-based scenarios and predictive forecasts of actual environmental trends. While the participant observed changes in group distributions, they were unable to explain why individual pollutant values did not update dynamically, pointing to an unclear conceptual model of how the scenario logic operated.

Overall, the participant's feedback demonstrated that the system's primary usability barriers stem from low interaction discoverability and ambiguous visual encodings. Among these, the baseline comparison ambiguity identified in E2 was prioritized for redesign, as it directly undermined the user's ability to accurately evaluate and reason about the visual evidence in Task 2.

C. Design Improvement

We selected the Task 2 comparison baseline issue for improvement because it directly affected the interpretation of the monthly attribute profiles shown in Figs. 7–9.

The explanatory text, colour legend and hover tooltip were revised to clarify that the baseline represents the median of all visible hours in the corresponding month, including the selected group. Measurement units were also added to the tooltip.

The changes were implemented in index.html and app.js. Only the interface descriptions were modified; the underlying data, calculations and visual encodings remained unchanged.

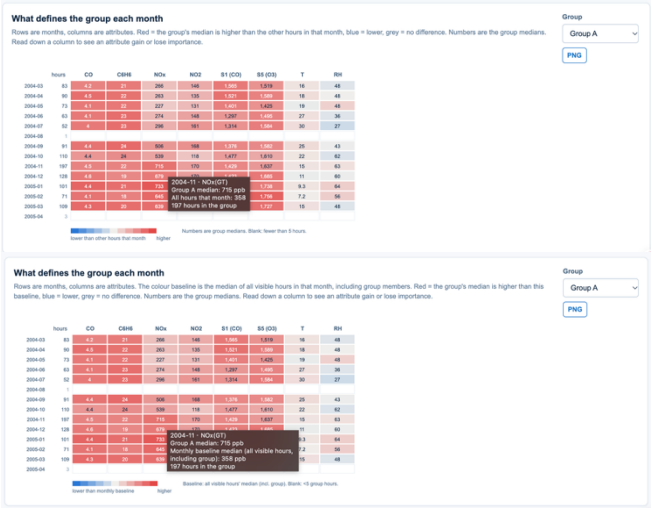


Fig. 13. Task 2 monthly attribute profile before and after the labelling improvement.

D. Verification and Limitations

Technical checks confirmed that the updated labels were consistent with the existing calculations. For Group A in November 2004, the NOx median remained 715 ppb, compared with the monthly baseline of 358 ppb. The positions, colours and values of the visual elements were unchanged.

The improvement makes the comparison baseline more explicit without altering the analytical findings in Task 2. However, no human usability retest was conducted, so improvements in user understanding or task performance have not been measured.

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